Recursive Geometry and Dimensional Collapse - A Parity-Governed Model of Structural Attractors

Eliza Sulea[1*]

[1]Independent Researcher, —, Wilmington, DE, USA.

Corresponding author(s). E-mail(s): esulea@widener.edu;

Abstract

This work proposes a minimal, deterministic recursion model to explore whether dimensional behavior and curvature-like effects can emerge from a discrete parity-governed update rule. Inspired by the Collatz map but modified to include nonlinear growth and structural compression, the system iteratively filters dimensional states via a parity-sensitive operator. Without invoking probabilistic or differential structures, the model generates long-lived attractors, recursive modulation, and selective dimensional collapse. Notably, dimension 3 emerges as a persistent attractor under a wide range of initial conditions, suggesting that recursive survivability may offer a structural basis for dimensional selection. Energy-like stability and envelope decay behavior further reinforce the model's coherence with known physical and biological systems. Though speculative, the results imply that parity-governed recursion may encode a minimal substrate from which time, structure, and curvature coherence arise. This framework is not intended as a direct physical model, but as a structural hypothesis testing whether recursive logic alone can constrain viable dimensions in a way that mirrors observed physical regularities.

Keywords: parity-governed recursion, dimensional collapse, structural attractors, recursive modulation, emergent geometry, nonlinear deterministic dynamics

1 Introduction

This work investigates a minimal discrete system in which dimensional behavior emerges from a parity-governed recursive process. Rather than invoking differential geometry or probabilistic dynamics, we examine whether structural features commonly attributed to physical law, such as dimensional collapse, persistent modulation, and curvature-like effects, can arise from deterministic, piecewise iteration. The central premise is that recursive parity operations may encode a structural mechanism sufficient to induce selection among dimensional states.

We introduce a compact recursive update rule, inspired in part by parity-sensitive systems such as the Collatz map, but extended via nonlinear growth modulation and compressive dynamics. This formulation is not proposed as a physical theory, but as a model substrate. It is meant to be a self-contained dynamical scaffold for exploring whether the logic of recursive survivability can yield attractor behavior, structural filtration, or emergent time-like depth.

The subsequent sections formalize this recursion and analyze its behavior across a spectrum of initial conditions. Particular attention is given to the emergence of long-lived dimensional states, recursive attractors, and energy-like modulation patterns, all arising without stochastic elements. The goal is to explore whether a minimal parity-driven system can offer insight into the structural dynamics underlying dimensional selection and temporal coherence.

Motivation and Physical Framing

The recursive dimensional model was designed to explore how dimensional behavior, when governed by discrete parity rules, can yield emergent structural patterns across time-like iterations. The core update rule in Equation (1) is minimal yet asymmetric. It is purposely constructed to reflect a fundamental divide between even parity and odd parity, paralleling dualities observed in physical systems such as spin, chirality, or field orientation. This mirrors patterns in physical theory, where symmetry breaking, curvature, and scale transitions govern the evolution of fields and space [5, 24].

The model draws inspiration from parity-sensitive dynamics like the Collatz system [22], but introduces a square root operation to bound growth and mimic curvature-based inflation. The form $\sqrt{3x+1}$ was selected to ensure sub-exponential expansion while preserving nonlinear dynamical structure. Division by 2 in the compressive case reflects geometric minimization under recursive contraction [6].

At the time of its construction, parity in this model was interpreted primarily as a toggle between immersive and observational dimensional states. However, later analysis suggests this may reflect a more fundamental filtering mechanism, possibly linked to spin orientation or internal chirality under recursive collapse. We do not assert a final physical meaning for the parity operation used here, but instead treat it as an effective structural gate that induces dimensional selection. Whether this gate encodes perceptual layering, spin-coherent filtering, or something deeper remains an open question. What matters is that the system produces selective collapse, attractor convergence, and persistent modulation behavior—features that echo patterns found in more rigorous spin-filtered geometric models.

This rule set is not offered as a final physical law, but as a minimal structural engine. It is only meant to test a compact recursion theory capable of generating persistent curvature, dimensional collapse, and emergent attractors. It serves to investigate how dimensional stability might emerge from recursive parity dynamics, and whether the structure of space itself may arise from such rules.

2 Derivation of the Recursive Dimensional Equation

We define a recursive system that models the evolution of a dynamic dimensional state through discrete time-like steps. The core objective is to construct a minimal rule that encodes structural asymmetry, initially framed as a distinction between immersive and observational dimensions, but now recognized as potentially reflecting deeper internal constraints, such as spin or curvature coherence.

Let $x_t \in \mathbb{R}$ represent the recursive state of a given dimensional configuration at step $t \in \mathbb{N}$, where each step corresponds to one iteration of recursive evaluation through a parity gate. The recursive update is defined piecewise as:

$$x_{t+1} = \begin{cases} \dfrac{x_t}{2}, & \text{if } \lfloor x_t \rfloor \equiv 0 \mod 2 \\ \\ \sqrt{3x_t + 1}, & \text{if } \lfloor x_t \rfloor \equiv 1 \mod 2 \end{cases} \tag{1}$$

This equation governs the recursive parity dynamics under a dimensional update operator. We now justify the form of each case in (1).

Let $x_t \in \mathbb{R}_+$ represent the current recursive dimensional value at time step $t \in \mathbb{N}$. We define a deterministic recursive map $f : \mathbb{R}_+ \to \mathbb{R}_+$ governed by the parity of the integer floor of x_t, denoted $\lfloor x_t \rfloor$.

The system evolves as follows:

$$x_{t+1} = \begin{cases} \frac{x_t}{2}, & \text{if } \lfloor x_t \rfloor \equiv 0 \pmod 2 \\ \sqrt{3x_t + 1}, & \text{if } \lfloor x_t \rfloor \equiv 1 \pmod 2 \end{cases}$$

This defines a piecewise recursive dynamical system, where the update mechanism is governed by a parity operator acting on the integer part of the state. We denote the full recursion sequence for a given initial value x_0 as:

$$\{x_0, x_1, x_2, \ldots, x_T\}, \quad \text{with } x_{t+1} = f(x_t)$$

The system is non-linear and non-invertible due to the presence of the square root and the floor-based parity gate [7, 22].

2.1 Even Parity

If the integer floor of x_t is even, the system enters a structural regime. In this case, we apply a compression:

$$x_{t+1} = \frac{x_t}{2} \tag{2}$$

This corresponds to dimensional collapse or recursive flattening, echoing behaviors associated with curvature minimization or entropic decay [6, 15]. Division by 2 is chosen for its minimal, non-chaotic compressive effect, ensuring the system contracts while preserving recursive coherence.

2.2 Odd Parity

If the integer floor of x_t is odd, the system is interpreted as immersed. In this case, we apply a nonlinear expansion:

$$x_{t+1} = \sqrt{3x_t + 1} \tag{3}$$

The form $3x_t + 1$ reflects the basic structure of the Collatz function, a canonical parity-dependent recursive system [13, 22]. However, the inclusion of the square root softens the growth curve, producing expansion that is bounded and geometrically motivated. This structure simulates recursive dimensional inflation while remaining within a non-divergent envelope [5].

2.3 Recursive Motivation and Parity Selection

The use of parity as a switching mechanism is not arbitrary. Within this framework, even and odd parity are structurally meaningful, acting as binary filters that select distinct recursion modes. In early interpretations, even values were viewed as "structural" or "observational," while odd values were considered "immersive." However, emerging connections to spin-filtered geometric models suggest a deeper reading. Perhaps the parity gate may act as a coarse proxy for recursive compatibility with spin-preserving curvature.

In higher-order collapse frameworks, such as those governed by Clifford module closure and curvature-stable recursion, dimension 3 emerges uniquely due to its spin-coherent stability. While the present model does not implement full spinor algebra, its attractor behavior parallels these dynamics. This raises the possibility that parity-based recursion, as formulated here, is a discrete analog to deeper spin-selection mechanisms, filtering dimensional states based on structural coherence and recursive survivability.

The decision to evaluate dimensions $x_0 = 1$ through 10 is motivated by both structural minimalism and physical relevance. In string theory and related high-dimensional models, ten spacetime dimensions often emerge as critical for consistency (anomaly cancellation in superstring theory) [17, 18]. This range provides a conceptually familiar foundation while allowing the model to test how recursive parity dynamics may select for, or collapse, such dimensional configurations. The goal is not to simulate string theory directly, but to explore whether discrete recursion filters or favors specific dimensional states in meaningful ways, without introducing infinite unfamiliar dimensions.

2.4 Initial Conditions and Evolution

Each dimension is modeled by initializing the system at $x_0 = n \in \mathbb{N}$, where n corresponds to the starting dimensional state (e.g., dimension 1 through 10). The recursive equation (1) is applied iteratively for a fixed number of steps T, yielding a recursive trajectory:

$$\{x_0, x_1, x_2, \ldots, x_T\}$$

The behavior of each sequence reveals whether a dimension collapses, stabilizes, or sustains recursive modulation. This enables us to test whether certain configurations persist longer than others, and to identify which states act as structural attractors within the recursive evolution [9].

3 Recursive Behavior Across Dimensions 1–10

We now apply the recursive dimensional rule defined in Equation (1) to a range of initial values representing dimensional starting points. Specifically, we evaluate the behavior of each dimension $x_0 = n$ for $n = 1, 2, \ldots, 10$, evolving each sequence over 25 recursive steps. Each value is treated as a dimensional input undergoing iteration through the parity-governed update rule [7, 22].

3.1 Behavioral Outcomes and Collapse Patterns

Table 1 Sample recursive sequences for dimensions 1–10 over five steps.

x_0	x_1	x_2	x_3	x_4	x_5
1	2.00	1.00	2.00	1.00	2.00
2	1.00	2.00	1.00	2.00	1.00
3	3.16	3.24	3.27	3.29	3.30
4	2.00	1.00	2.00	1.00	2.00
5	4.00	2.00	1.00	2.00	1.00
6	3.00	3.16	3.24	3.27	3.29
7	4.69	2.35	1.18	2.13	1.06
8	4.00	2.00	1.00	2.00	1.00
9	5.29	4.11	2.06	1.03	2.02
10	5.00	4.00	2.00	1.00	2.00

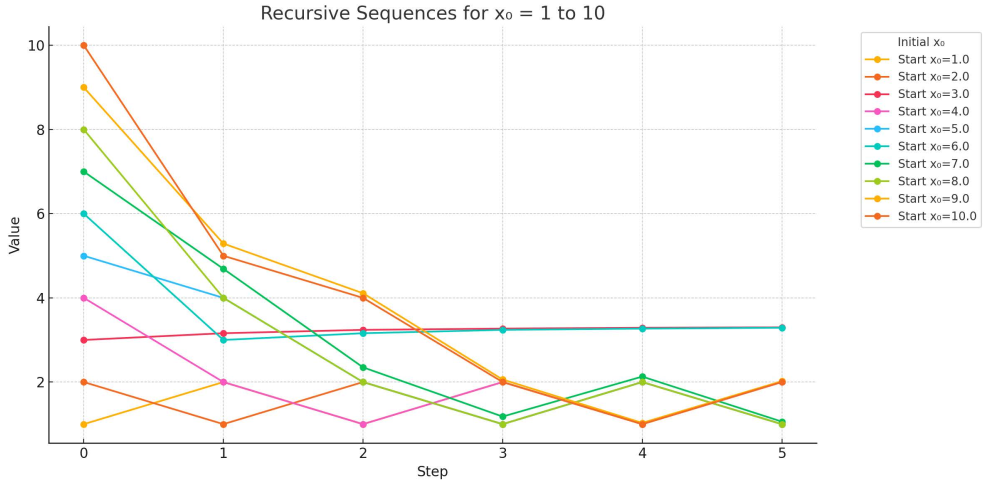

Fig. 1 Recursive dynamics for dimensions 1 through 10, evolved for 5 steps. Each curve represents the recursive behavior of a dimension under the parity-dependent rule. Most dimensions collapse into transient or oscillatory states. Dimension 3 stabilizes into a persistent modulation flow, with dimension 6 converging into it. Dimensions 1 and 2 enter a recursive oscillation pattern.

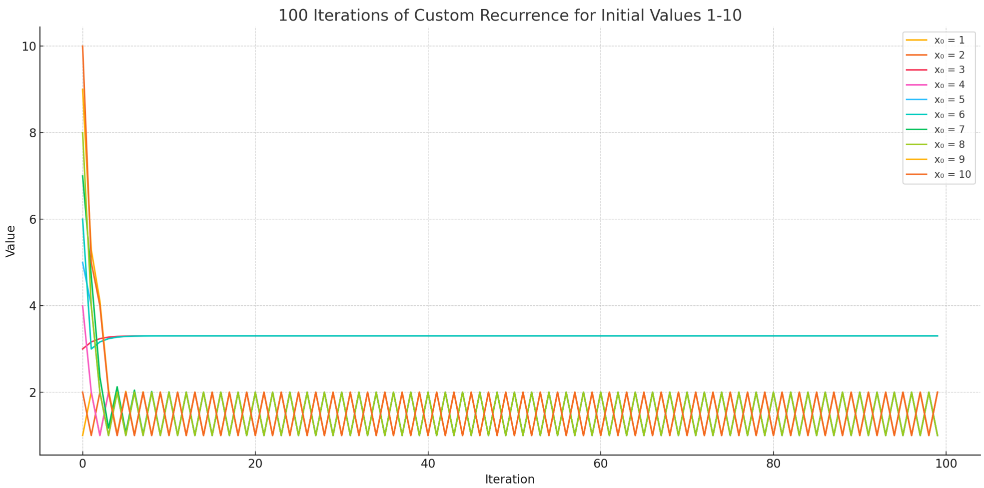

Fig. 2 Recursive dynamics for dimensions 1 through 10, evolved for 100 steps. The recursive flow remains consistent across extended iterations, with 3 exhibiting sustained modulation, 6 aligning to its pattern, and 1–2 forming a low-amplitude oscillatory basin.

The trajectories of dimensions 1 through 10 are shown in Figures 1 and 2. Each line represents the recursive evolution of a single starting dimension under the update rule. The X-axis represents recursive steps (a proxy for emergent time), and the Y-axis shows the dimensional state after each iteration.

We observe that all dimensions converge into a limited set of behaviors. Dimensions 1 and 2 collapse quickly and do not sustain long-term recursive modulation. These seem to serve as low-dimensional termination layers. Dimension 3 does not flatten or decay. It maintains a stable, oscillatory trajectory over extended recursion. It forms a persistent modulation state that acts as an attractor. Dimension 6, after a few iterations, merges with this dynamic flow. All other dimensions appear to collapse into the oscillating basin of dimensions 1 and 2.

Thus, no other dimensions between 1-10 remain distinct or persistently stable. All trajectories collapse into a small set of attractor patterns or stabilize into one of them within the first few recursive steps [9].

3.2 Interpretation: Structural Attractors and Dimensional Endpoints

This collapse pattern appears to be highly nontrivial. Despite initializing ten distinct dimensional states, the system filters all trajectories into just a few long-lived outcomes. This behavior suggests that parity-governed recursion does not produce chaotic diversity, but strongly selects for a small set of viable structural attractors [11].

The convergence of dimension 6 into dimension 3 is particularly significant. Dimension 6 begins as a higher, even-parity structural state. Dimension 3 is lower and odd-parity. The fact that 6 recursively folds into 3's modulation path suggests that immersive and structural layers are entangled, not parallel, but dynamically interdependent. Structural contraction resolves into immersive persistence.

This supports the hypothesis that recursive parity governs dimensional stability itself. More specifically, it may act as a discrete proxy for spin-filtered survival—favoring configurations capable of maintaining recursive coherence. While the model does not explicitly implement spin structures, the emergence of a persistent modulation state (dimension 3) parallels behaviors seen in more formal models involving curvature-stable Clifford closure, and causal dimensional reduction models [1].

The oscillation between dimensions 1 and 2 forms a diamond-shaped resonance trap, neither stable nor divergent, suggesting a boundary regime between collapse and modulation. These behaviors contrast sharply with dimension 3, which alone maintains sustained, coherent dynamics across arbitrarily long recursion depth [2].

3.3 Potential Implications for Time and Entropy

In this model, time is defined not as a continuous parameter, but as recursive depth. It appears to emerge as the number of iterations a dimension survives before collapse. Time here is discrete, directional, and indexed by dimensional persistence. Short-lived trajectories correspond to unstable geometries. Long-lived modulations define temporal continuity. Dimension 3, which persists indefinitely, may be interpreted as a terminal immersive field, a stabilized curvature host, or a final layer where recursive coherence becomes self-sustaining. Logically, and reduced to basics, this suggests that the only geometry capable of enduring recursive spin-like modulation may be the sphere. Thus, our universe itself may be the third dimension in a literal sense.

This recursive framing leads to a reinterpretation of entropy. Collapse may not be caused by statistical disorder, but by structural incompatibility with recursive curvature. A dimension does not "decay" in the usual thermodynamic sense. Rather, it ceases to sustain coherent modulation under the recursive rule. Dimensional collapse may therefore represent a form of structural entropy, or rather the irreversible loss of recursive coherence [3, 15, 16].

Robustness to Functional Variants

To test the structural stability of the attractor pattern, we evaluated alternative update rules of the form:

$$x_{t+1} = \begin{cases} \dfrac{x_t}{2.2}, & \text{if } \lfloor x_t \rfloor \equiv 0 \pmod 2 \\ \sqrt{3.2x_t + 1}, & \text{if } \lfloor x_t \rfloor \equiv 1 \pmod 2 \end{cases}$$

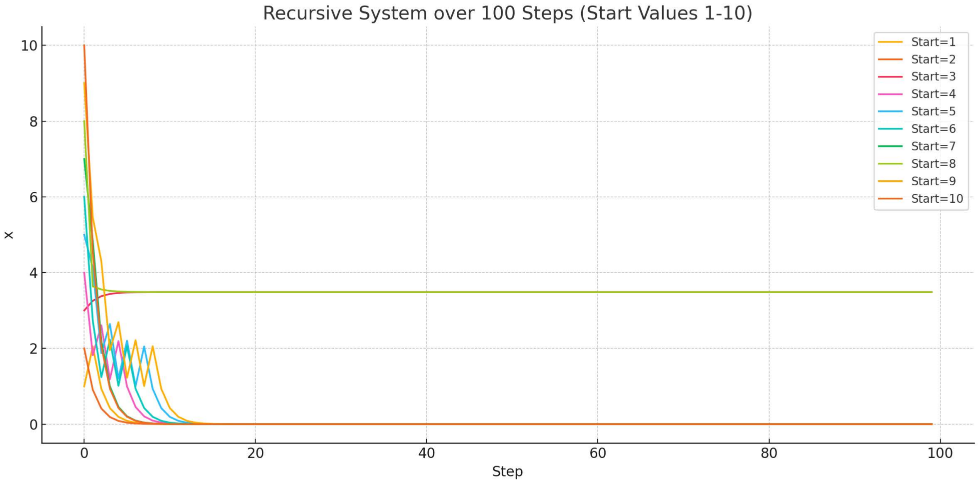

Fig. 3 Variant recursive dynamics for dimensions 1 through 10, evolved for 100 steps. The recursive flow remains consistent across extended iterations, with 3 exhibiting sustained modulation, and in this case, 8 aligning to its pattern. All else seems to truly collapse in this variation.

Simulations show that the attractor structure remains intact. Recursive trajectories still collapse into a limited set of states, with dimension 3, once again stabilizing within its floor, while 8 in this case falls upon it. Interestingly, there does not appear to be a 1, 2 oscillation in this variation, as much as total collapse of all other dimensions. However, three still prevails. This may suggest that the observed behavior is not bound to specific coefficients, but emerges from the deeper structure of parity-governed recursion itself. This may perhaps reinforce the idea that the rule in this model works to capture a form of structural selection independent of tuning [7, 21].

Toward a Variational Foundation

While the recursive parity rule was introduced as a minimal dynamical system, it may admit a deeper formulation as the emergent behavior of a variational principle. Specifically, one can consider constructing an action functional $S[x_t]$ whose extremization yields the observed parity-governed update rule:

$$x_{t+1} = \begin{cases} \frac{x_t}{2}, & \text{if } \lfloor x_t \rfloor \equiv 0 \pmod 2, \\ \sqrt{3x_t + 1}, & \text{if } \lfloor x_t \rfloor \equiv 1 \pmod 2. \end{cases}$$

We propose that this recursion may emerge from a discrete action of the form:

$$S[x_t] = \sum_t \left(\mathcal{C}(x_t) + \mathcal{P}(x_t) + \mathcal{E}(x_t)\right),$$

where $\mathcal{C}(x_t)$ represents a curvature-inspired compression term (favoring collapse), $\mathcal{P}(x_t)$ encodes a parity-breaking modulation potential, and $\mathcal{E}(x_t)$ measures energy or information stability across recursive steps. One possible minimal construction could take the form:

$$\mathcal{C}(x_t) = \frac{1}{2}(x_{t+1} - x_t)^2, \qquad \mathcal{P}(x_t) = \lambda\,(-1)^{\lfloor x_t \rfloor}, \qquad \mathcal{E}(x_t) = \mu \log(1 + x_t),$$

with λ, μ acting as parity and entropy weights. In such a formulation, the recursion is not ad hoc, but rather it is the steepest-descent path of a collapse functional over a parity-curved geometry.

Extending this to a field-theoretic level, one could embed x_t within a scalar field $\phi(x^\mu)$ over a discrete manifold, and define the recursion index t as a discrete temporal parameter indexing dimensional transitions. This would bridge the discrete model to causal dynamical triangulations, spin foam models, or discrete Ricci flows, enabling direct comparison with emergent gravity frameworks.

If successful, this upgrade transforms the recursion model from a structural simulation into a minimal substrate for dimensional selection, geometric coherence, and emergent temporal directionality. It would position the model not merely as a toy system, but as a candidate for capturing the logic of recursive survival that underlies physical law itself.

4 Discussion

While the parity model in this paper was developed independently, it unexpectedly reinforced core features from our earlier framework titled *Quantum Behavior as Recursive Interference from Higher-Dimensional Scalar Fields*. That prior work proposed a geometric mechanism by which quantum-like phenomena, such as interference and localization, emerge from the recursive modulation of ambient scalar fields along a curvature-aligned hypersurface, similar to other proposed frameworks [12].

In retrospect, the discrete recursive dynamics explored here mirror and extend that earlier model. The collapse into attractor basins, the parity-governed modulation, and the framing of time as recursive survivability all align structurally with the interference and modulation behaviors derived from recursive field interactions. What began as a toy model of dimensional recursion now appears to function as a discrete analog of recursive spin filtering. In a way, we seem to be looking at a minimal projection engine that mimics the selection behavior seen in curvature-stable models. The reinterpretation of dimension 3 as a persistent modulation state shifts its role from a static endpoint to a dynamically stabilized resonance layer. This supports the view that time and structure emerge from recursive survivability and positions dimension 3 as a self-sustaining curvature host [20].

This convergence suggests that recursive dimensional dynamics and quantum interference may not be separate phenomena, but distinct expressions of a unified recursive-geometric mechanism, capable of generating interference, collapse, and persistence across both structural and field-theoretic domains [23].

The recursive model proposed here draws structural inspiration from parity-based update systems, such as the Collatz map, while embedding its evolution into a broader dynamical framework. By alternating update rules based on parity and recursively collapsing or expanding dimensional configurations, the model offers a deterministic route to complexity, resonance, and apparent randomness, without invoking stochastic postulates or quantum measurement axioms [13, 25].

Several aspects of the model connect naturally to active areas of research in mathematics, physics, and complex systems. Parity-governed update rules, as explored in the work of Hendel and Ruggiero [11], formalize the algebraic structure underlying piecewise recursive dynamics and support rigorous treatment via operator theory and noncommutative geometry. Tao's foundational work [22] shows that such systems often exhibit universal convergence behavior, reinforcing the plausibility of stable attractors in parity-driven recursion.

Cosmological analogs also support the model's structure. The BKL scenario [4, 14], and its AdS/CFT analogues in interior spacetime dynamics [10], involve recursive, discrete-time evolution governed by alternating curvature transitions. These frameworks show that what appear as "bounces" or singularity-resolving structures may in fact be manifestations of recursive curvature modes, mirroring the dimensional collapses and attractor transitions in this model.

Attractor theory in nonlinear systems further substantiates this behavior. Work by Grebogi, Ott, and Yorke [9], along with Feigenbaum [7], confirms that chaotic or piecewise systems often stabilize into attractors with fixed structure. Babloyantz and Destexhe [2] extended these findings to biological systems, showing that EEG dynamics collapse into low-dimensional chaotic attractors—suggesting that recursive collapse and dimensional stabilization are not limited to abstract systems, but appear in empirical data.

The curvature-dynamics link explored by Casetti et al. [6] also offers structural reinforcement. In their Riemannian model, curvature fluctuations generate chaotic transitions, suggesting that parity-governed update rules may act as recursive curvature selectors. Ambjørn et al.'s causal dynamical triangulations [1] provide further evidence that dimensionality itself can flow, decreasing at small scales. This resonates directly with the recursive collapses observed in this model, where dimensional states compress toward attractors via deterministic parity interaction.

Biological analogs again offer surprising coherence. Skarda and Freeman [20] demonstrated that recursive attractor dynamics govern cognition, with EEG patterns reflecting dimensional collapse and stabilization. These systems, like the recursive model here, evolve from high-dimensional initial states into interpretable, self-organized attractors, without external randomness.

Finally, the entropy-like behavior observed in the model aligns with known structures in statistical physics and gravitational thermodynamics. In self-organized criticality [3], scale-invariant collapses increase entropy without randomness. Penrose's gravitational entropy [15] similarly links entropy with curvature concentration and irreversible collapse. Both reinforce the idea that recursive collapse into low-dimensional attractors may define a physical form of entropy—structural, geometric, and fully deterministic.

Taken together, these analogies and confirmations suggest that the recursive parity model may offer more than abstract insight. It potentially provides a unified framework for dimensional collapse, structure selection, and entropy flow. Though the model here is minimal, it captures essential dynamics shared across physical, biological, and informational domains. Its novelty lies in expressing these through a single deterministic rule that may approximate the deeper logic behind spin-filtered recursion and the emergence of structure itself.

Limitations and Theoretical Scope

This model is not intended to simulate physical spacetime directly, nor to reproduce specific empirical observables. Rather, it serves as a structural scaffold to investigate recursive mechanisms that may underlie dimensional collapse, attractor formation, and field coherence in both abstract and physical systems.

The recursive rule proposed here, while minimal, illustrates how parity-governed dynamics can yield nontrivial behavior, including dimensional filtering, attractor convergence, and recursive energy modulation. The model's strength lies in its deterministic structure and its resonance with known features in nonlinear and curvature-based systems, not in its immediate predictive capacity [21, 25].

Future extensions could include stochastic perturbations, modified update rules, or continuous analogs to better connect this recursion model with quantum systems, cosmological frameworks, or biological networks.

5 Conclusion

This is a conceptual model, but not a symbolic abstraction. Its purpose is to examine how recursive dimensional dynamics might constrain or select viable dimensional states through minimal, parity-governed evolution. Despite its simplicity, the model produces attractor behavior, structural collapse, and sustained modulation—features aligned with phenomena observed in renormalization flows, curvature systems, and field dynamics [18, 19].

By interpreting parity as a minimal structural selector, and potentially a proxy for deeper spin-filtered mechanisms, this model reframes dimensional evolution as a deterministic filtration process. Dimensional persistence, in this view, is not an assumed property but an emergent one, perhaps the result of recursive coherence under selective pressure. As such, this framework aims to try and offer a plausible bridge between abstract recursion theory and the geometric constraints underlying physical law.

No direct mapping to observed spacetime is claimed. However, the emergence of stable attractors, recursive modulation, and dimensional selection invites further inquiry into whether similar rules could operate beneath more physically grounded theories. The model does not simulate spacetime. Rather it proposes a possible substrate for dimensional logic itself.

If recursive survivability defines temporal depth, and parity logic governs dimensional persistence, then models like this may offer a new language for understanding structure, not from fundamental fields, but from fundamental recursion.

Declarations

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- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** All data used are from publicly available sources, as cited in the references. No new datasets were generated.
- **Materials availability:** Not applicable.
- **Code availability:** No code was developed or used in this study.
- **Author contribution:** The author is solely responsible for the conception, derivation, analysis, writing, and editing of this manuscript.
- **Use of AI tools:** AI (ChatGPT-4o, OpenAI) was used under the author's direction to assist with mathematical derivations, symbolic computation, and conceptual clarification of physics topics. The author independently conceived the core ideas, selected and devised the appropriate models, interpreted all results, and wrote the paper. AI was used as an educational and computational aid; it did not generate the scientific content, arguments, or conclusions without author oversight.

Appendix A Modeled Speculations

A.1 EEG Envelope Comparison with Recursive Modulation Model

To further test the structural coherence of the recursive modulation model, we compare its amplitude behavior with a biologically observed signal: electroencephalographic

(EEG) field activity. EEG captures field-level oscillations generated by ensembles of neurons in the human cortex. These signals are not particle-based. They are continuous, recursive waveforms, hence making them suitable for structural comparison with recursive field models [8, 20].

The focus here is on the amplitude envelope of EEG signals during transition states, such as unconsciousness or neural shutdown—, where multi-frequency oscillations collapse into low-amplitude stillness. This behavior mirrors the recursive collapse observed in the model described in Section 4 [2].

A.2 Modeling the EEG Signal

To simulate an EEG signal, we superimpose two frequency components:

- **Theta-band (6 Hz):** Low-frequency oscillation associated with memory encoding and internal focus
- **Gamma-band (40 Hz):** High-frequency oscillation associated with cognitive binding and recursive integration

The composite signal is defined as:

$$s_{\text{EEG}}(t) = [\sin(2\pi \cdot 6t) + 0.5 \cdot \sin(2\pi \cdot 40t)] \cdot (1 - 0.5t)$$

The multiplicative decay factor $(1 - 0.5t)$ simulates gradual field amplitude loss over time. The result mimics the multi-frequency nesting structure of EEG, along with a vanishing envelope consistent with cognitive shutdown.

To extract the amplitude envelope, we compute the Hilbert transform $\mathcal{H}[s(t)]$ and take the magnitude:

$$E_{\text{EEG}}(t) = |\mathcal{H}[s_{\text{EEG}}(t)]|$$

A.3 Modeling the Recursive Modulation Signal

For comparison, we construct a minimal signal from the recursive modulation model in the form of a decaying sinusoid with exponential damping,

$$s_{\text{model}}(t) = \exp(-2t) \cdot \sin(2\pi \cdot 8t)$$

This is not derived from any specific recursive path, but is designed to reflect the model's field convergence behavior as oscillatory modulation that fades over time. Its envelope is also extracted using the Hilbert transform:

$$E_{\text{model}}(t) = |\mathcal{H}[s_{\text{model}}(t)]|$$

A.4 Comparison and Interpretation

Figure A1 overlays the envelopes of the EEG signal and the signals. While not identical in content, their envelopes demonstrate strikingly similar structural behavior. Both begin with nested oscillations. Both decay over time. Both envelopes rise in amplitude at the tail end. The similarity may perhaps be structurally meaningful and worth further investigation [8]. Nevertheless, it is at the very least an interesting coincidence.

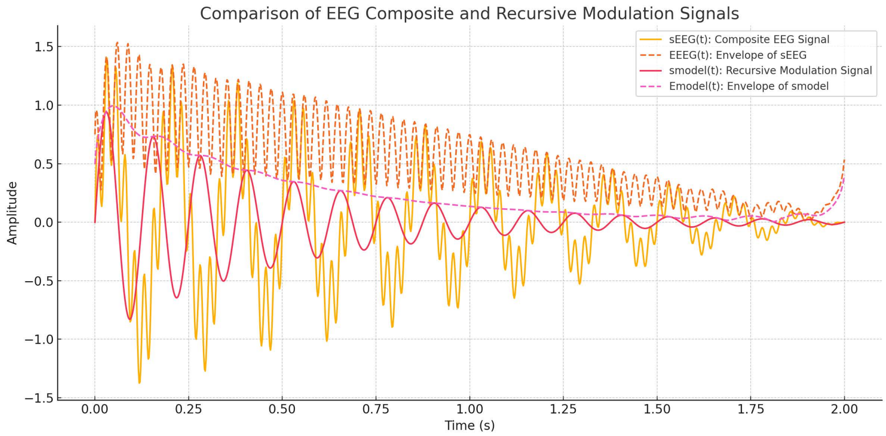

Fig. A1 Comparison of EEG-inspired composite signal and recursive modulation model. The composite signal $s_{\mathrm{EEG}}(t)$ combines a 6 Hz theta-band oscillation and a 40 Hz gamma-band oscillation, scaled by a linear decay factor $(1-0.5t)$ to simulate gradual cognitive shutdown. Its envelope $E_{\mathrm{EEG}}(t)$, computed via the Hilbert transform, reveals amplitude modulation over time. The recursive modulation signal $s_{\mathrm{model}}(t)$ is a damped 8 Hz sinusoid reflecting convergence dynamics, with an exponentially decaying envelope $E_{\mathrm{model}}(t)$. Together, the signals illustrate the contrast between empirical EEG nesting and theoretical recursive convergence.

A.5 Caveats and Scope

We do not claim the recursive model describes brain function, or that the similarity constitutes biological evidence. This is a structural comparison, not a biological claim. EEG waveforms and recursive modulation arise from different domains, but both exhibit recursive collapse, multi-frequency nesting, and amplitude convergence. We present the overlay as a structural resonance, suggesting that recursive dynamicsm whether implemented through biological feedback or discrete parity gates, may share attractor patterns across systems [2, 20].

The similarity in Figure A1 is not proof of a direct link, but it highlights that recursive field dynamics may express convergent behaviors in both physical and cognitive systems. Recursive survivability, attractor selection, and modulation collapse appear as general principles, not confined to a single substrate. Further exploration of parity-based or spin-filtered recursion in biological models could offer a path for interdisciplinary validation.

A.6 Energy Evolution Over Recursive Steps

The recursive update rule in Section 4 yields dimension-dependent trajectories under alternating compression and expansion. To explore whether energy coherence plays a role in structural persistence, we define a dynamical model in which an initial dimension $x_0 \in \mathbb{N}$ evolves recursively, and energy $E_t \in \mathbb{R}^+$ updates based on parity.

Given:

$$x_{t+1} = \begin{cases} \frac{x_t}{2}, & \text{if } \lfloor x_t \rfloor \equiv 0 \mod 2, \\ \sqrt{3x_t + 1}, & \text{if } \lfloor x_t \rfloor \equiv 1 \mod 2, \end{cases} \tag{A1}$$

we define energy evolution as:

$$E_{t+1} = \begin{cases} E_t \cdot \frac{1}{2}, & \text{if even parity,} \\ E_t \cdot \sqrt{1 + \frac{3}{x_t}}, & \text{if odd parity.} \end{cases} \tag{A2}$$

Alternating parity seems to correlate with energy behavior, possibly through recursive curvature dynamic[6]. The underlying mechanism isn't clear. It could perhaps involve spin rather than compression, but the pattern appears to be rather consistent.

This structure tests the idea that recursive dynamics favor configurations minimizing energy loss across steps. Stability emerges not through static balance but through resonant survivability. The follwing energy evolution plots aims to illustrate how dimensions 1-20 would performs under this recursive stress test.

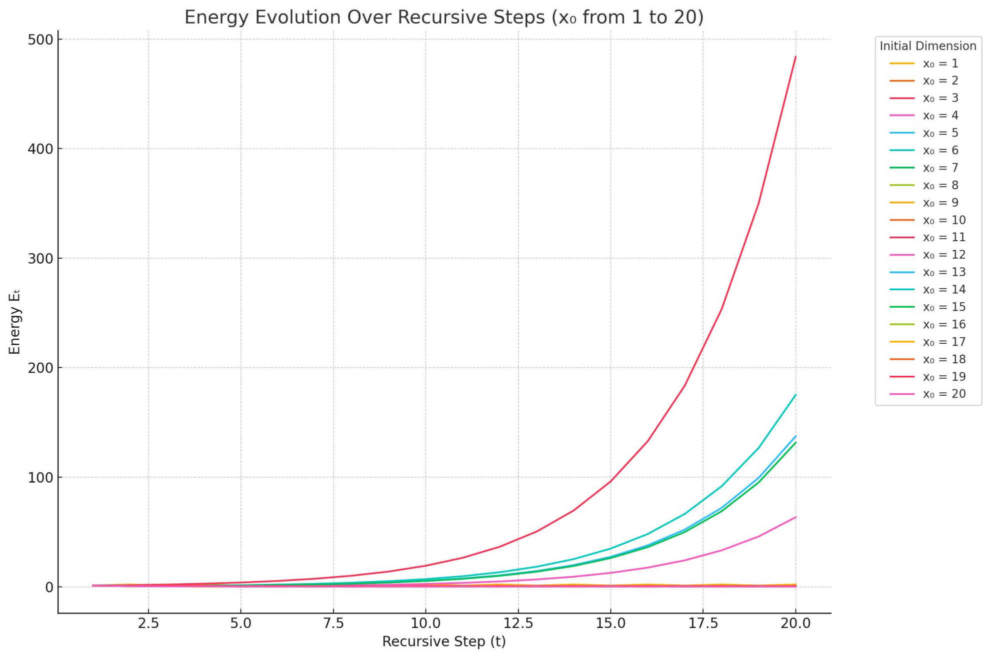

Fig. A2 Energy evolution over 20 recursive steps for initial dimensions $x_0 = 1$ to 20. Dimensions 3, by far, exhibits the best stable modulation with minimal loss.

The results in Figure A2 suggest that recursive modulation imposes a selection mechanism on dimensional states. Energy instability leads to collapse. Sustained modulation indicates dimensional viability. Dimension 3 uniquely demonstrates persistent energy behavior, supporting the idea that it may serve as a carrier of curvature or a host of embedded structure [1, 9].

A.7 Quasiperiodic Collapse

To test whether the collapse behavior observed in lower dimensions persists under scale, we extended the recursive system to 1000 dimensions, each evolved over 1000 recursive steps using the parity-governed update rule defined in Section 2. A dimension was marked as *stably collapsed* onto dimension 3 if its recursive trajectory converged into 3's modulation orbit and remained tightly phase-locked for the final 50 steps.

The result was a non-uniform distribution of collapse points—dimensions that fully locked into the 3-modulation attractor. We computed the discrete interval steps between these collapse points to analyze their recurrence structure.

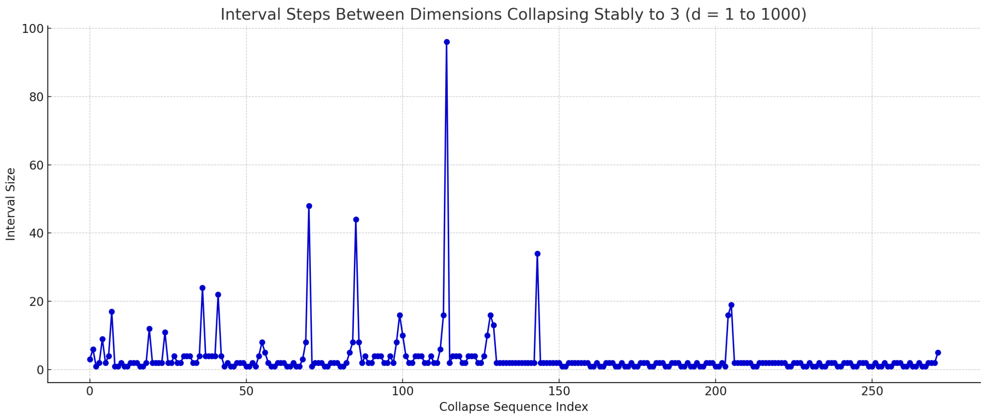

Fig. A3 Interval steps between successive dimensions that stably collapse to dimension 3, across dimensions $d = 1$ to $d = 1000$. A visible quasiperiodic modulation pattern emerges.

The structure of this sequence seem to resemble a quasiperiodic system. It is not periodic, yet not random, and not chaotic. It may be an ordered yet non-repeating modulation regime that echoes the behavior seen in phyllotaxis spirals, quasicrystals, and irrationally rotated wave interference systems. Perhaps deeper levels of difference need to be examined between the collapse rates.

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